

Structure of a DBMS

➤ Introduction

- Structure of DBMS shows the internal working components of a database system. It explains how data is stored, managed, and accessed.
- DBMS acts as an interface between users and database. It controls storage, retrieval, and security of data.
- Different components work together to manage database operations. Each component has a specific function.

➤ Main Components of DBMS Structure

1. Query Processor
2. Storage Manager
3. Database Engine
4. Transaction Manager
5. File Manager
6. Buffer Manager
7. Authorization & Integrity Manager
8. Data Files
9. Data Dictionary

➤ Query Processor

- Query processor interprets and executes SQL queries. It converts user queries into low-level instructions.
- It checks syntax and optimizes queries. It helps in faster query execution.

➤ Functions

- Parsing queries
- Query optimization
- Query execution

2. Storage Manager

- Storage manager controls data storage in disk memory. It acts as interface between database and file system.
- It manages files, indexes, and records. It ensures efficient storage and retrieval.

➤ Functions

- Store data physically
- Manage indexes
- Retrieve records

➤ 3. Database Engine

- Database engine is the core service of DBMS. It processes and manages database operations.
- It allows users to create, update, and delete data. It controls overall database activities.

➤ 4. Transaction Manager

- Transaction manager handles database transactions safely. It maintains ACID properties.
- It controls concurrency and recovery. It prevents data inconsistency.

➤ Functions

- Commit transactions
- Rollback operations
- Concurrency control

➤ 5. File Manager

- File manager manages database files on storage devices. It organizes records inside files.

- It controls allocation of disk space.
It handles file access methods.

➤ **6. Buffer Manager**

- Buffer manager manages temporary data in main memory.
It reduces disk access time.
- Frequently used data is stored in buffer memory.
This improves system performance.

➤ **7. Authorization & Integrity Manager**

- This component provides security to database.
It controls user permissions.
- It also checks integrity constraints.
It ensures valid and consistent data.

➤ **Functions**

- User authentication
- Access control
- Constraint checking

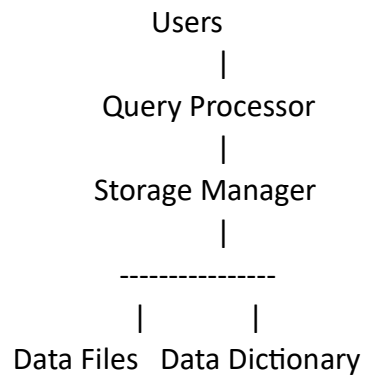
➤ **8. Data Files**

- Data files store actual database information.
They are stored on secondary storage devices.
- Tables, records, and indexes are stored here.
DBMS accesses these files during operations.

➤ **Data Dictionary**

- Data dictionary stores metadata about database.
Metadata means “data about data.”
- It contains table names, attributes, and constraints.
DBMS uses it for management purposes.

Simple DBMS Structure Diagram



Advantages of DBMS Structure

- Provides organized data management.
- Improves security and consistency.
- Supports fast data retrieval.
- Reduces redundancy.
- Supports multiple users.

Disadvantages

- Structure is complex.
- Requires more memory and cost.